Personal Portfolio Website

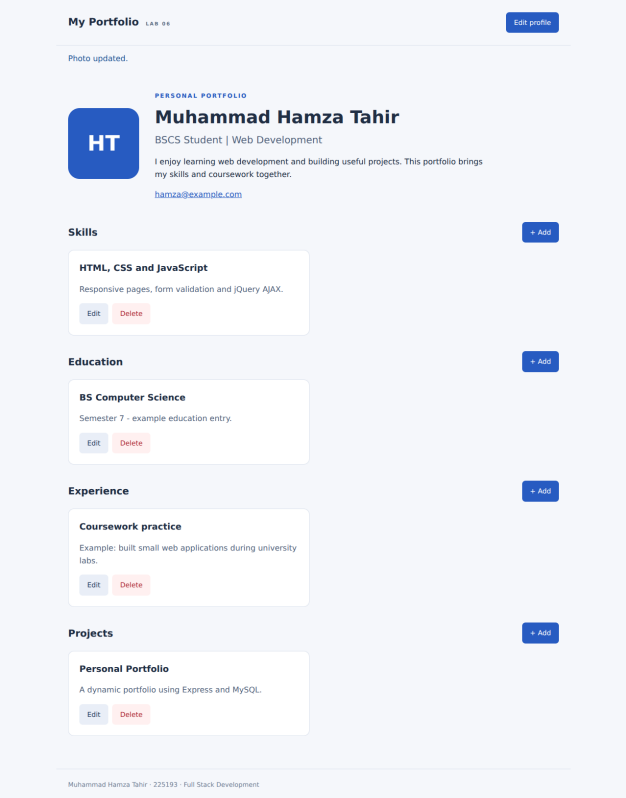
Full Stack Development - Lab 06 | CS-301L

Name: Muhammad Hamza Tahir
Registration ID: 225193
Semester: 7

What the website does

This is a simple personal portfolio. On the first visit, the user can enter a profile. The website also lets the user add, edit and delete skills, education, experience and projects. A profile image can be uploaded, updated or removed.

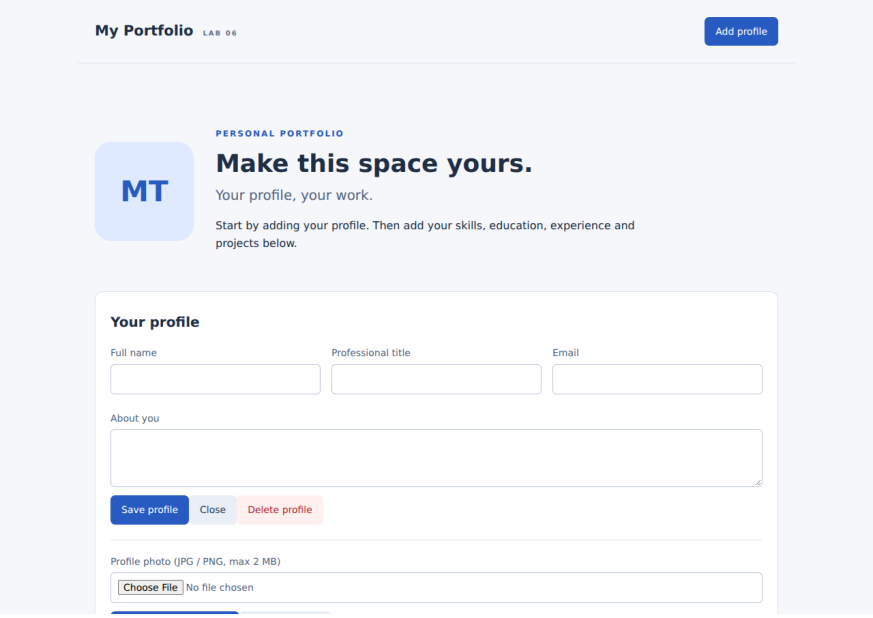
HTML and CSS provide the page layout. jQuery AJAX sends requests to an Express server. The server reads and writes data in MySQL. Uploaded images are kept in the uploads folder, and their paths are saved in the database.



Screenshot: populated portfolio. Education, experience and project text are examples used for testing; they are not verified personal history.

1. Frontend and AJAX

The first-open form collects a name, title, email and bio. Each portfolio section uses the same card layout and Add, Edit and Delete controls. AJAX refreshes the displayed data without reloading the whole page.




```
$.ajax({
  url: '/api/profile',
  method: 'PUT',
  contentType: 'application/json',
  data: JSON.stringify(body)
}).done(function () {
  load();
});
```

This excerpt shows the AJAX save pattern used in public/app.js. The actual application also displays a message when a request fails.

2. Backend and image upload

Express handles the API endpoints. The backend checks required text fields, a valid email address, allowed categories and entry IDs. Parameterized queries pass user values to MySQL separately from SQL commands.

```
await pool.execute(
  'INSERT INTO entries (category,title,detail) VALUES (?,?,?)',
  [category, title, detail]
);

await pool.execute(
  'UPDATE entries SET category=?,title=?,detail=? WHERE id=?',
  [category, title, detail, id]
);
```

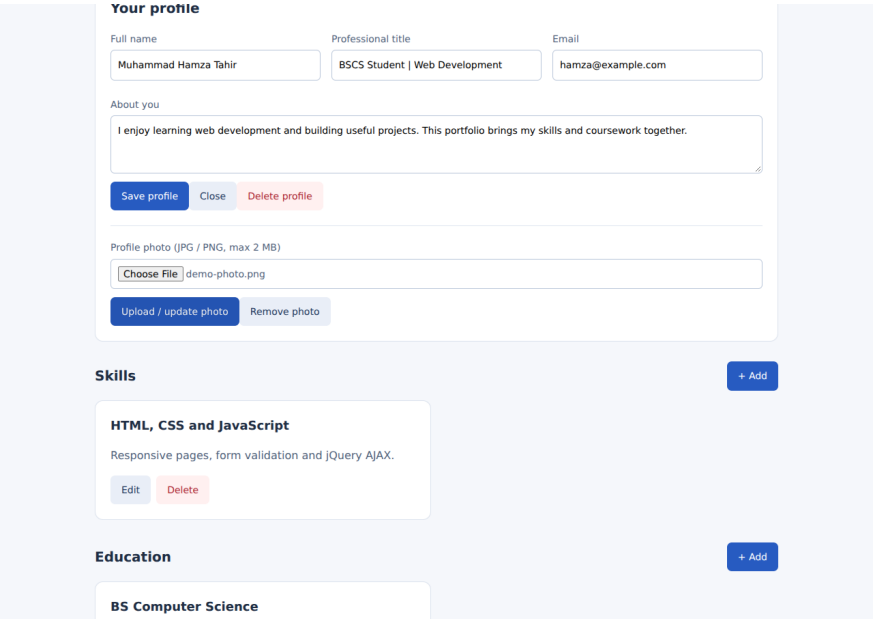


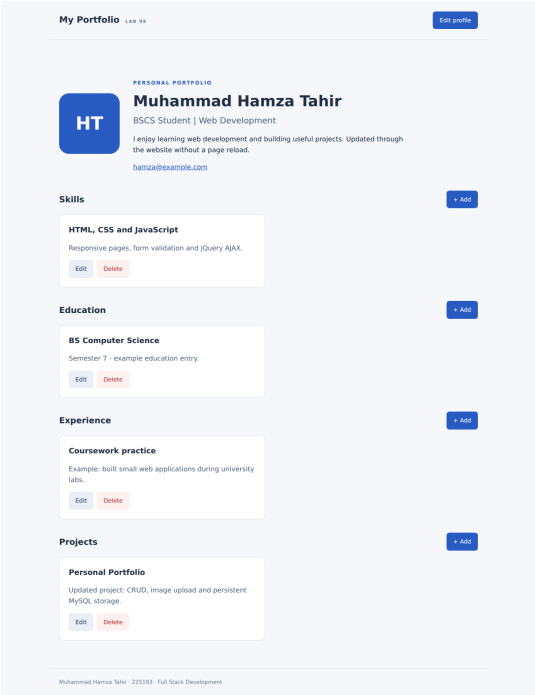
Image upload uses a multipart AJAX request and Multer. JPG and PNG signatures are checked, and the file size is limited to 2 MB. Each saved image has a unique filename. Updating or removing a photo also removes the old file.

The initials image in this screenshot is a demonstration image, not a personal photograph.

3. Database and persistence

The profile table stores one profile (id = 1). The entries table stores the four section types. Import database.sql before running the server. The SQL file starts with empty tables so the first visit shows the profile form.

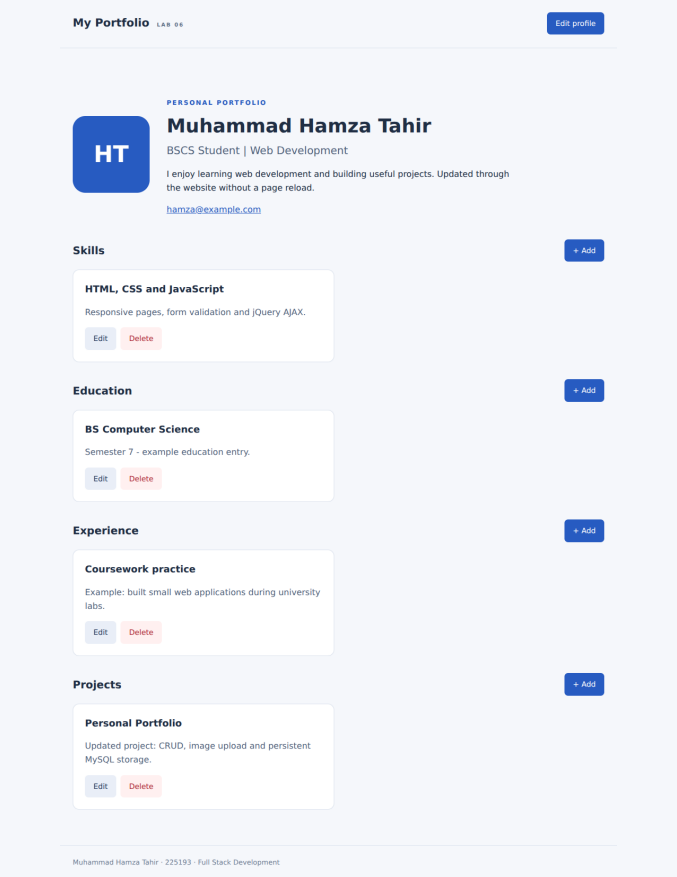
```
CREATE TABLE IF NOT EXISTS entries (
  id INT AUTO_INCREMENT PRIMARY KEY,
  category ENUM('skills','education','experience','projects')
    NOT NULL,
  title VARCHAR(120) NOT NULL,
  detail TEXT NOT NULL
);
```



Screenshot: data loaded after refreshing and restarting Node. The checks used MariaDB 10.11, a MySQL-compatible server used by XAMPP, with the provided SQL schema and mysql2 driver.

Main endpoints: GET /api/portfolio; PUT and DELETE /api/profile; POST and DELETE /api/profile/image; POST /api/entries; PUT and DELETE /api/entries/:id.

Responsive layout and checks



On a narrow screen, the profile and cards stack into one column. The mobile layout was checked at a width of 390 pixels without horizontal overflow.

Browser checks covered profile creation, editing and deletion; adding all four section types; editing and deleting entries; photo upload, replacement and removal; refresh and Node restart persistence; invalid profile rejection; and a missing entry response. No browser JavaScript errors were reported.

Run and demonstrate

1. Start MySQL and import database.sql.
2. Run npm install in the extracted project folder.
3. Copy .env.example to .env and set the database port and credentials.
4. Run npm start and open http://localhost:3000.
5. Enter your own details and demonstrate the add, edit, delete and image features.

README.txt contains the setup steps. The included MP4 shows the working browser with example content. RECORDING_GUIDE.txt explains how to record a fresh demo on Windows.

The handout lists one portfolio task but mentions three questions in the PDF requirement. This document covers that task in three implementation parts. This is a local single-user lab application; authentication is outside its scope.